Network Forensics Incident Report: NetSupport RAT Infection

Nov 6, 2026 · @Katelyn Craig

Executive Summary

Windows client DESKTOP-TEYQ2NR (10.2.28.88), used by Becka Rolf (account brolf), was infected with the NetSupport Manager remote access trojan (RAT) on 2026-02-28. The host first contacted the attacker's command-and-control (C2) server, 45.131.214.85, at 19:55:51 UTC. It then checked in about once a minute for the rest of the capture.

The infection was confirmed from network traffic alone. The C2 traffic carries the User-Agent "NetSupport Manager/1.3" and travels as unencrypted HTTP over TCP port 443, a port normally used for encrypted HTTPS. The host should be isolated and reimaged, and the brolf account's password reset.

Evidence

One packet capture was analyzed. Its SHA-256 hash was recorded before analysis to verify evidence integrity.

Item	Detail
File	2026-02-28-traffic-analysis-exercise.pcap
Source	Malware-Traffic-Analysis.net, "Easy As 123" exercise
SHA-256	3dc470f5490ec15b4024c513e87c67e0b17caeae2a2ce3d83541be24ef407e61
Packets	15,512
Tools	Wireshark (macOS), shasum (macOS Terminal)

```
3dc470f5490ec15b4024c513e87c67e0b17caeae2a2ce3d83541be24ef407e61
```

Figure 1. SHA-256 hash of the capture, computed with shasum -a 256 in macOS Terminal before analysis.

Network Environment

The infected host sits on a small Active Directory network, as described in the exercise scenario.

Item	Value
LAN segment	10.2.28.0/24
Domain	easyas123.tech
Active Directory name	EASYAS123
Domain controller	10.2.28.2

Infected Host Details

Each identifier was confirmed in at least one specific packet, listed below.

Item	Value	Evidence (frame)
IP address	10.2.28.88	2569: first connection to the C2 server
MAC address	00:19:d1:b2:4d:ad (Intel)	2569: Ethernet II source address
Hostname	DESKTOP-TEYQ2NR	233: Browser Request Announcement; 243: Kerberos AS-REQ, NetBIOS address
User account	brolf	243: Kerberos AS-REQ, CNameString
Full name	Becka Rolf	339: SAMR QueryUserInfo response from the domain controller

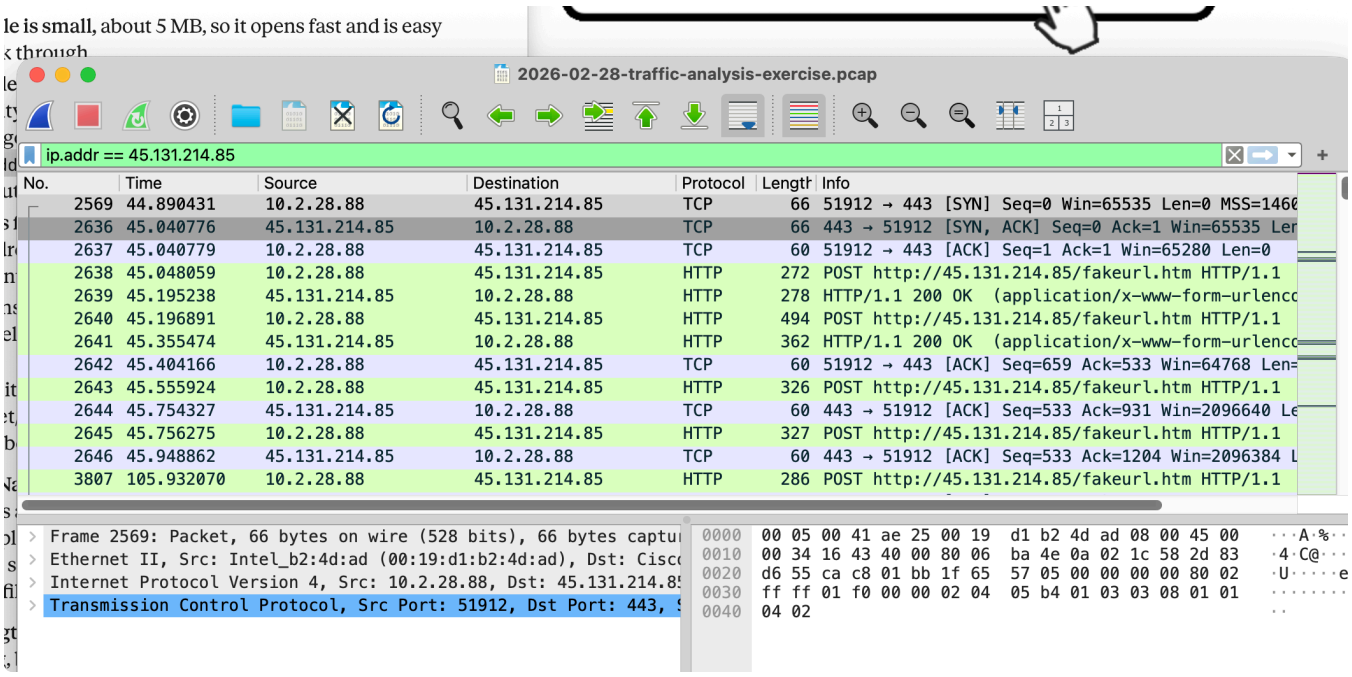


Figure 2. Filtering on the C2 address 45.131.214.85 identifies the victim, 10.2.28.88. Frame 2569's Ethernet II header shows its MAC address, 00:19:d1:b2:4d:ad.

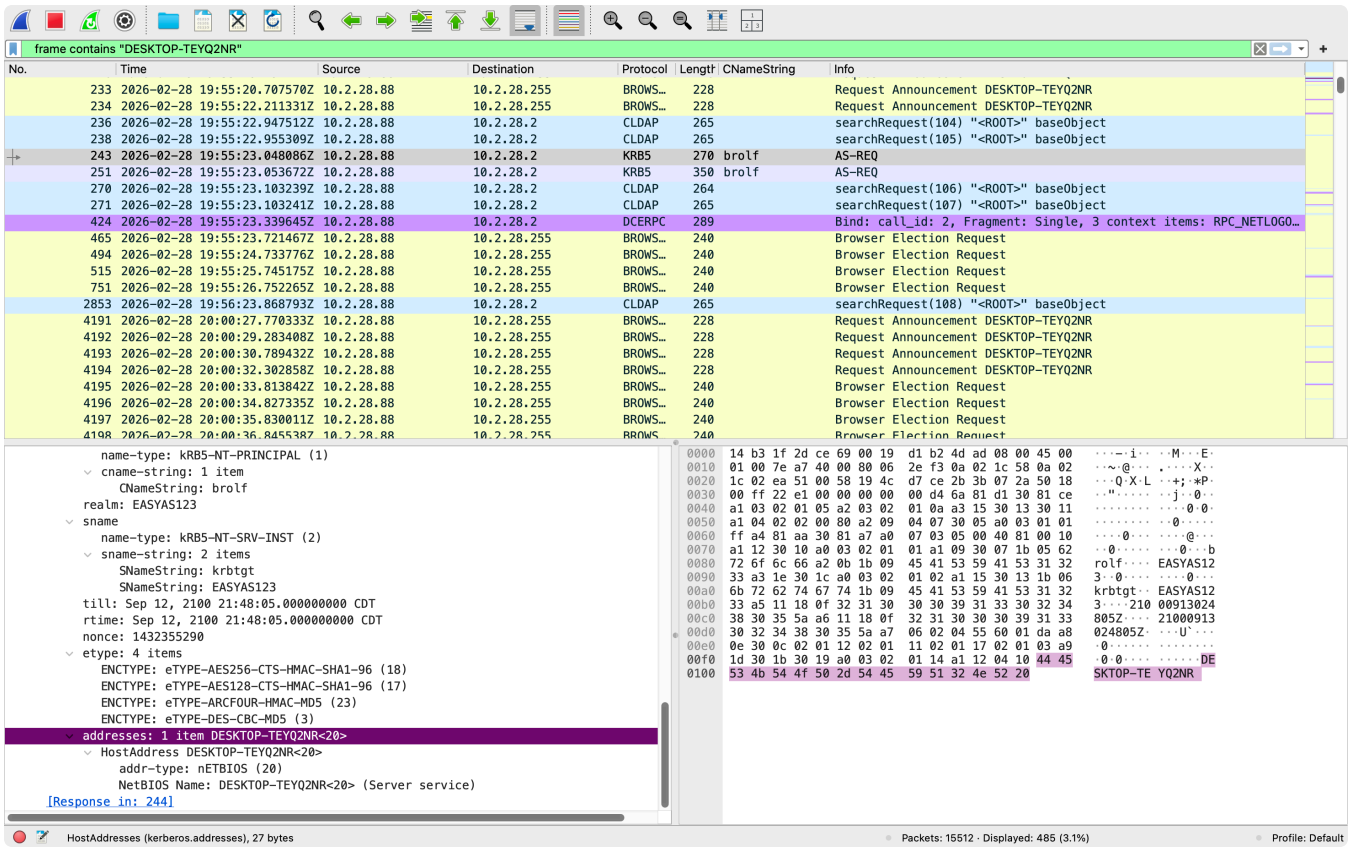


Figure 3. The hostname DESKTOP-TEYQ2NR in Browser announcements (frames 233, 234) and in frame 243's Kerberos addresses field.

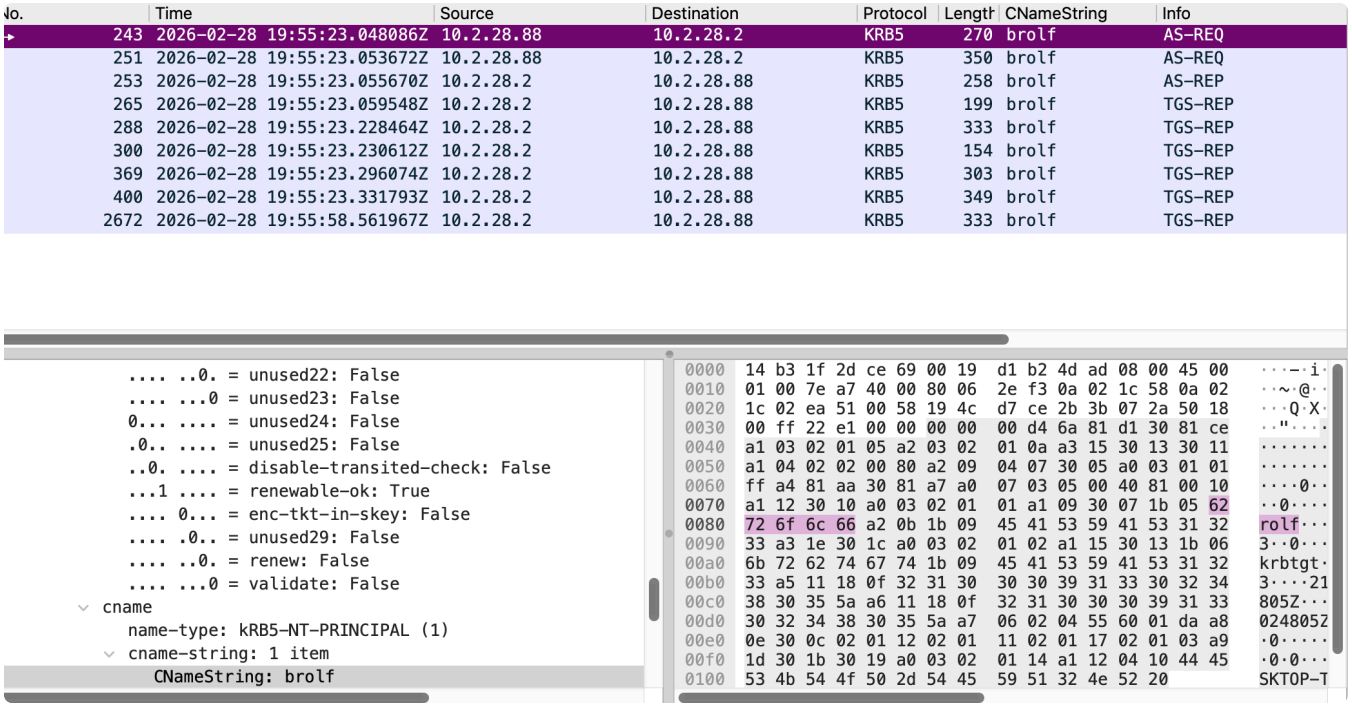
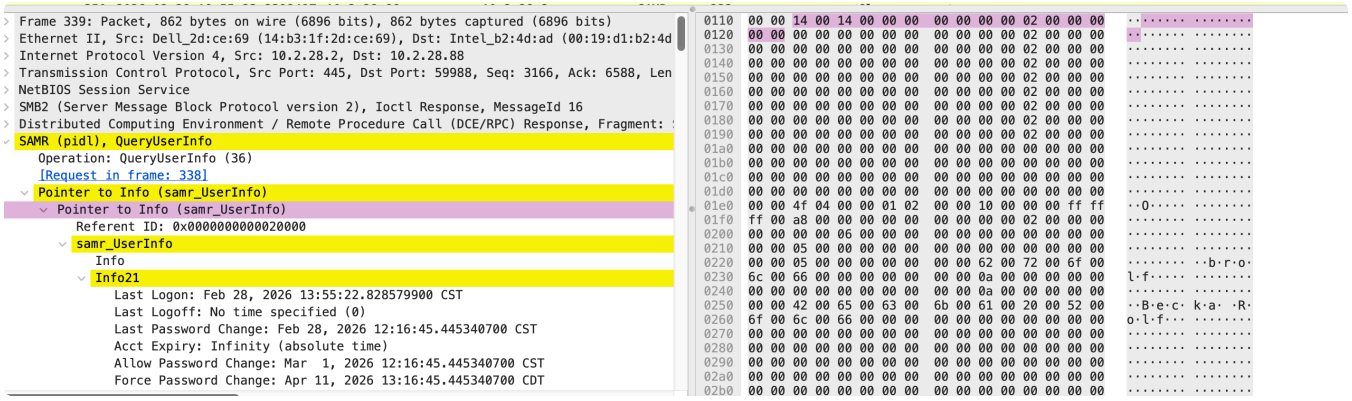


Figure 4. Kerberos traffic between the host and the domain controller shows the user account brolf (frame 243).



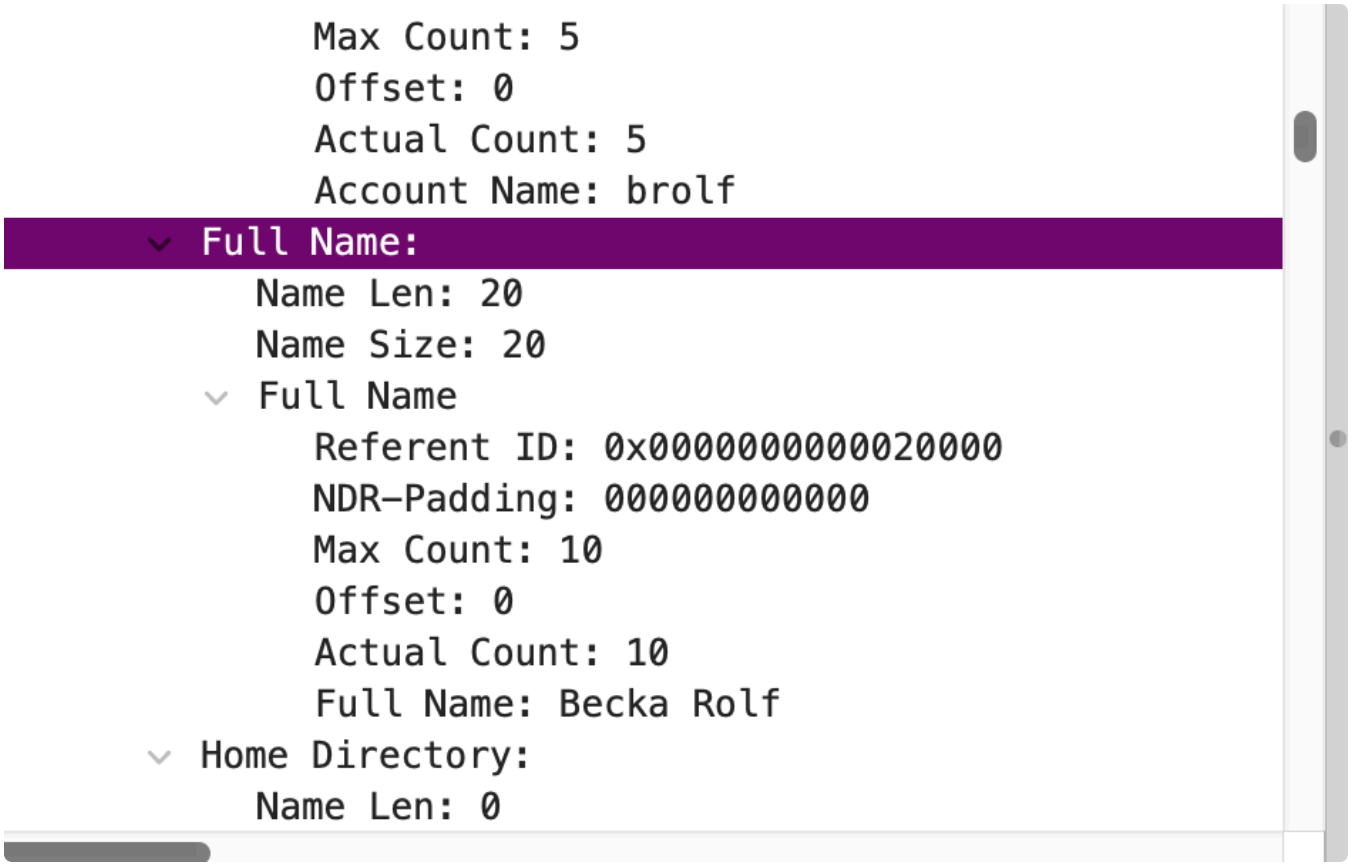


Figure 5. The domain controller's SAMR QueryUserInfo response (frame 339) gives the full name for brolf: Becka Rolf.

Methodology

The capture was analyzed in Wireshark, with display times set to UTC. Each step used a display filter to isolate the relevant traffic.

1. Hashed the pcap with SHA-256 before opening it, to record the evidence's original state.
2. Filtered on the alerted C2 address to identify the internal host talking to it: `ip.addr == 45.131.214.85`.
3. Read the victim's MAC address from the Ethernet II header of its outbound packets.
4. Found the Windows user account in Kerberos authentication traffic: `kerberos.CNameString && ip.addr == 10.2.28.88`.
5. Found the hostname with `frame contains "DESKTOP-TEYQ2NR"`, then confirmed it in Kerberos and Browser announcement packets.
6. Found the account holder's full name in SAMR traffic between the host and the domain controller: `samr && ip.addr == 10.2.28.88`.

7. Inspected the HTTP POST requests to the C2 server to confirm the malware family from the User-Agent header.

Findings and Indicators of Compromise

The host's traffic to 45.131.214.85 matches NetSupport Manager RAT command-and-control activity. Three details confirm it:

- **Malware identified by name.** The HTTP POST requests carry the User-Agent "NetSupport Manager/1.3" (frame 2640).
- **Unencrypted traffic on an encrypted port.** The C2 traffic is plain HTTP sent to TCP port 443, which normally carries encrypted HTTPS. This is a common way for malware to blend in with ordinary web traffic.
- **Regular beaconing.** After the first exchange, the host sent a POST to the same URL about every 60 seconds (frames 3807, 4093, 4134, 4149, 4201), a typical pattern for a RAT waiting for commands.

Indicator	Type	Note
45.131.214.85	C2 IP address	Contacted over TCP 443
http://45.131.214.85/fakeurl.htm	C2 URL	Target of every POST check-in
NetSupport Manager/1.3	HTTP User-Agent	Identifies the RAT
CMD=ENCD, ES=1, DATA=…	POST body fields	Encoded C2 data, form-urlencoded

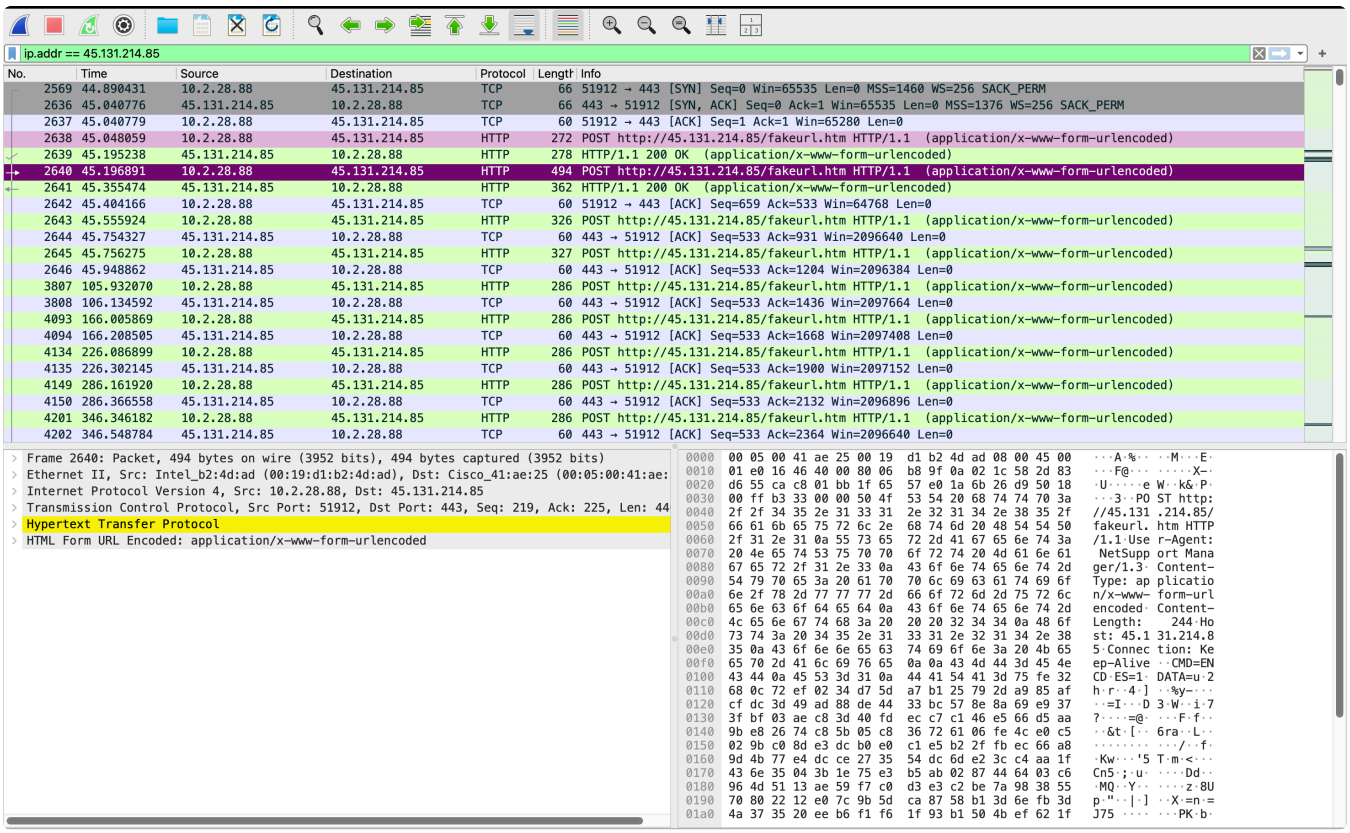


Figure 6. Frame 2640: an HTTP POST to /fakeurl.htm with the User-Agent "NetSupport Manager/1.3", followed by check-ins about every 60 seconds.

Timeline (UTC, 2026-02-28)

The user logged on about 30 seconds before the host first contacted the C2 server.

Time (UTC)	Frame	Event
19:55:20	233	DESKTOP-TEYQ2NR announces itself on the network
19:55:23	243	brolf authenticates to the domain controller (Kerberos AS-REQ)
19:55:23	339	Domain controller returns account details for brolf, full name Becka Rolf (SAMR)
19:55:51	2569	First connection from 10.2.28.88 to C2 server 45.131.214.85:443
19:55:51	2638	First HTTP POST to /fakeurl.htm

Time (UTC)	Frame	Event
19:56:52	3807	First regular check-in; repeats about every 60 seconds
20:00:52	4201	Last check-in reviewed in this analysis

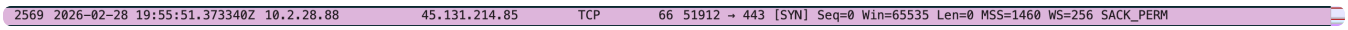


Figure 7. First contact with the C2 server: frame 2569 at 2026-02-28 19:55:51 UTC.

Conclusion and Lessons Learned

DESKTOP-TEYQ2NR (10.2.28.88), used by Becka Rolf (brolf), is infected with NetSupport Manager RAT and is in regular contact with the attacker at 45.131.214.85. Recommended actions:

- Isolate the host from the network and reimage it.
- Reset the brolf account's password and review its recent activity.
- Block 45.131.214.85 at the firewall and search other hosts' traffic for the same indicators.

Lessons learned:

- **Start from the alert.** Filtering on the alerted C2 address identified the victim immediately; the rest of the investigation built outward from that one host.
- **The full name was not where expected.** LDAP traffic held no readable name. It was found in SAMR traffic, where Windows asks the domain controller for account details. Next time, check SAMR early.
- **Confirm each finding twice.** The hostname appeared in both Browser announcements and Kerberos traffic, which makes the finding stronger than a single packet.

Sources

- Malware-Traffic-Analysis.net, 2026-02-28 traffic analysis exercise: Easy As 123